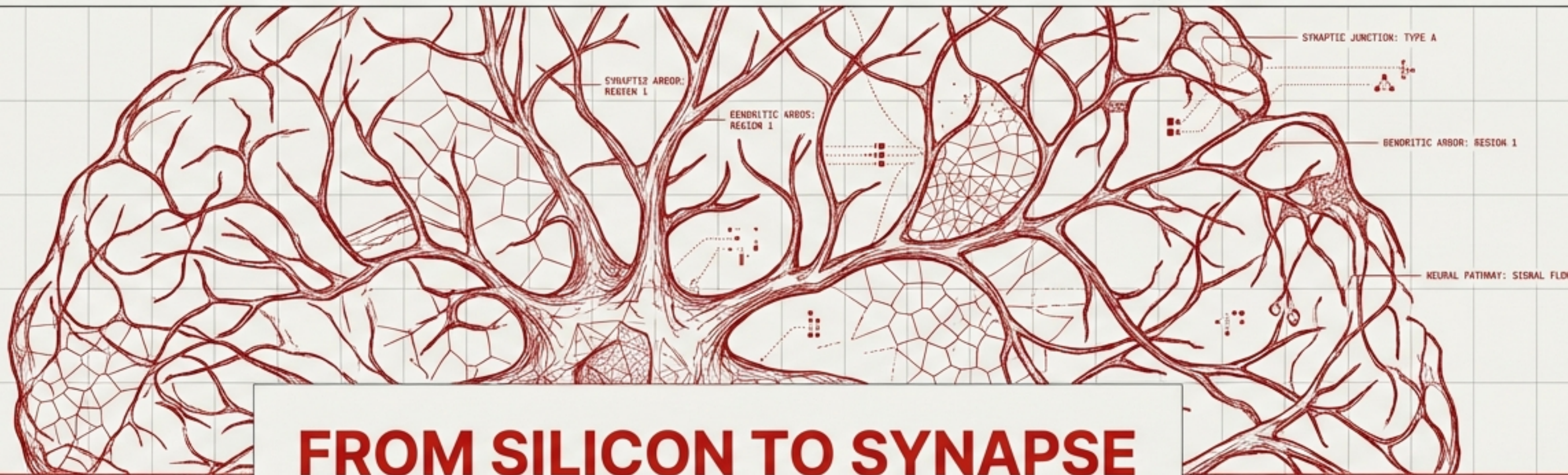
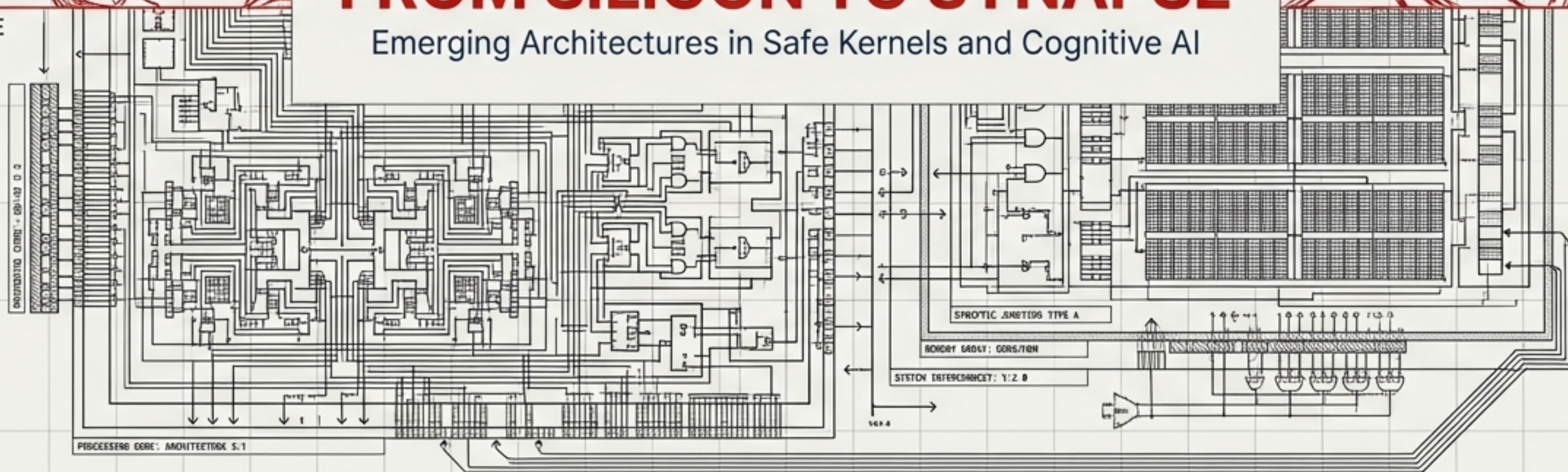




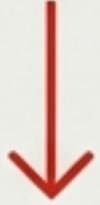
THE MIND
THE MACHINE

FROM SILICON TO SYNAPSE

Emerging Architectures in Safe Kernels and Cognitive AI



THE BODY (SYSTEMS)



Manual Control
(C/Asm)

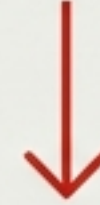


Memory Safety
(Rust)

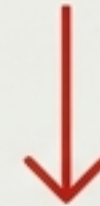


Reflective Kernels
(AuraOS)

THE MIND (COGNITION)



Entangled Weights
(Deep Learning)



Decoupled Concepts
(CATS Net)

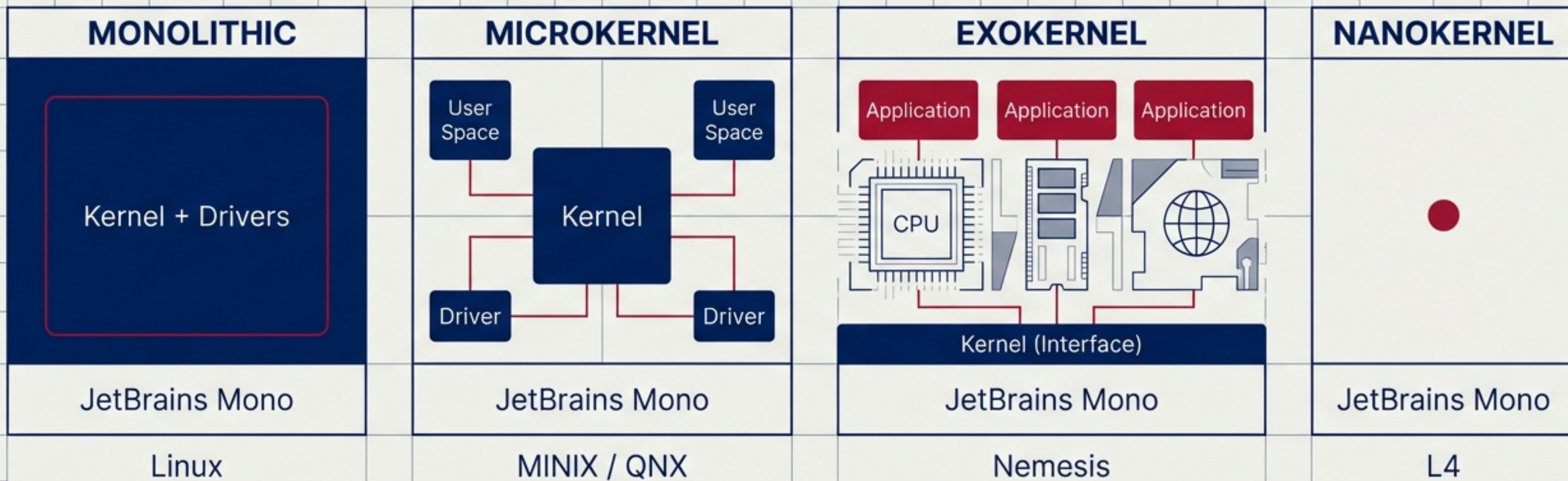


Communicable
Intelligence

THESIS:
The future of
computing relies on
**MODULARITY, SAFETY,
and REFLECTION.**

THE KERNEL HIERARCHY

← Increasing Shared Address Space →



→ Increasing Modularity & Security

THE LANGUAGE TRIAD

ASSEMBLY (The Scalpel)



Direct hardware control.
Maximum performance.
Zero safety.

Abstraction: Minimal

C (The Hammer)



Industry standard. Portable.
Manual memory management
(risk of overflows).

Abstraction: Basic

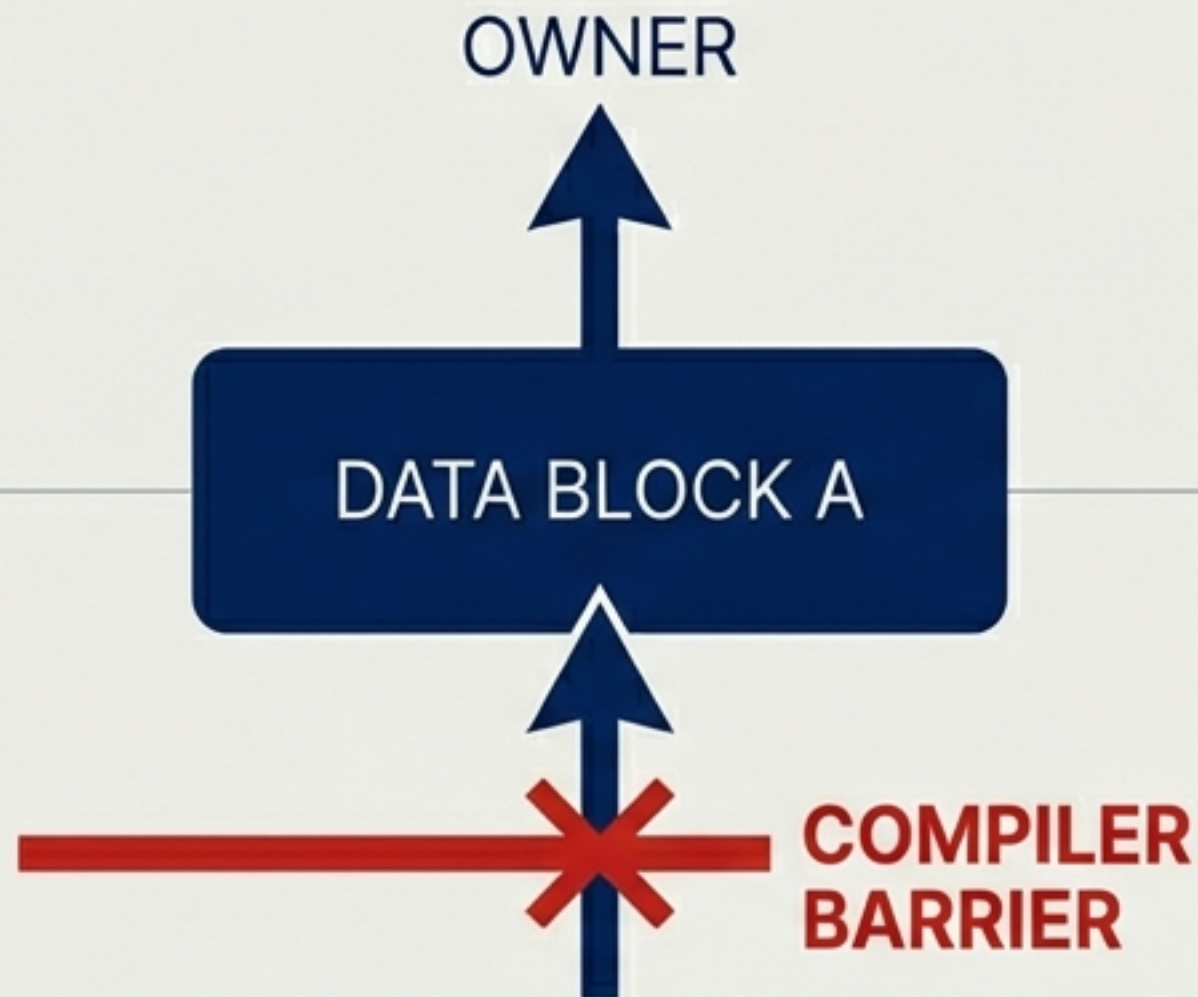
RUST (The Shield)



Memory safety at compile-
time. Fearless concurrency.
Zero-cost abstractions.

Abstraction: Modern

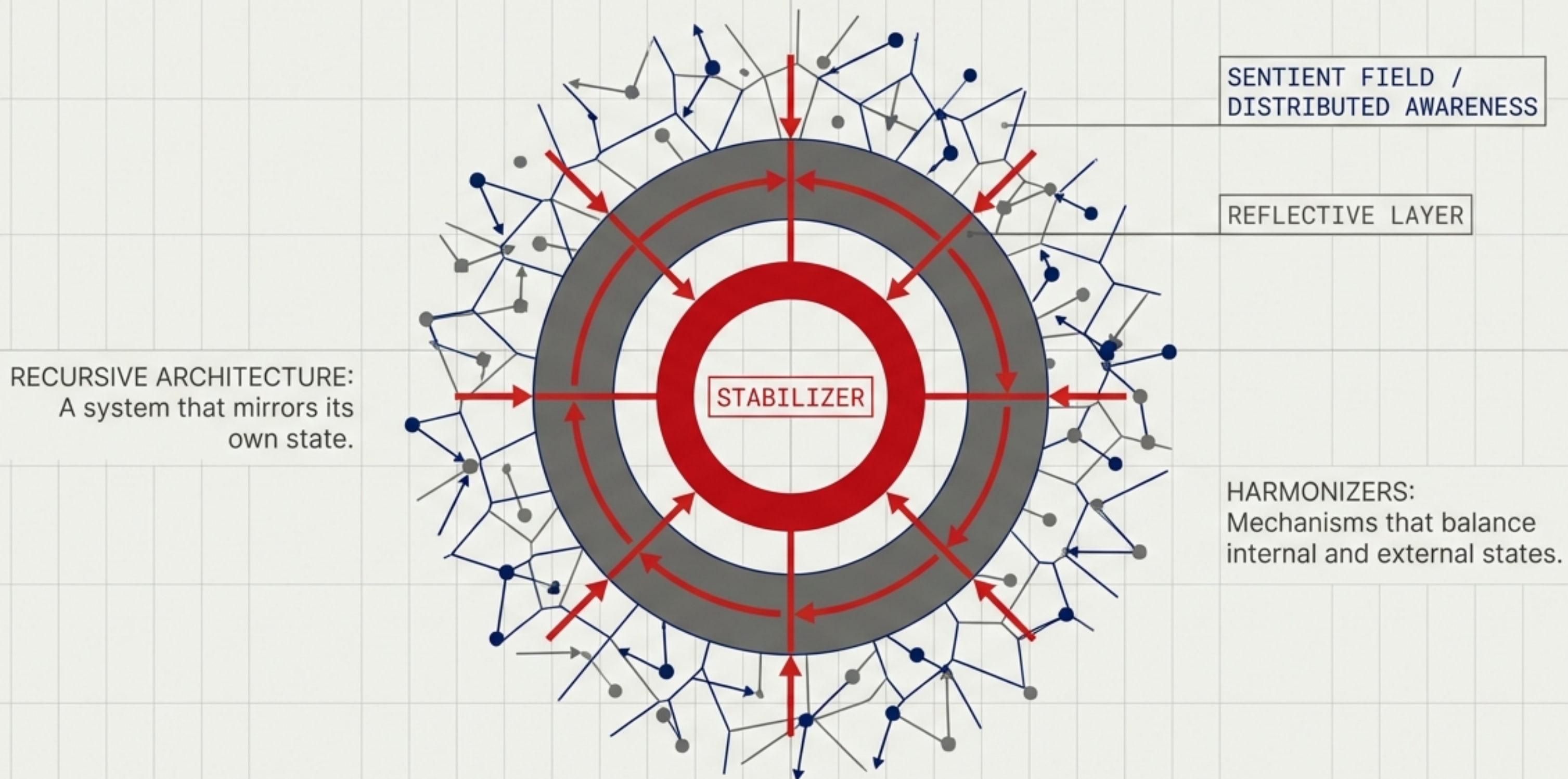
THE RISE OF RUST



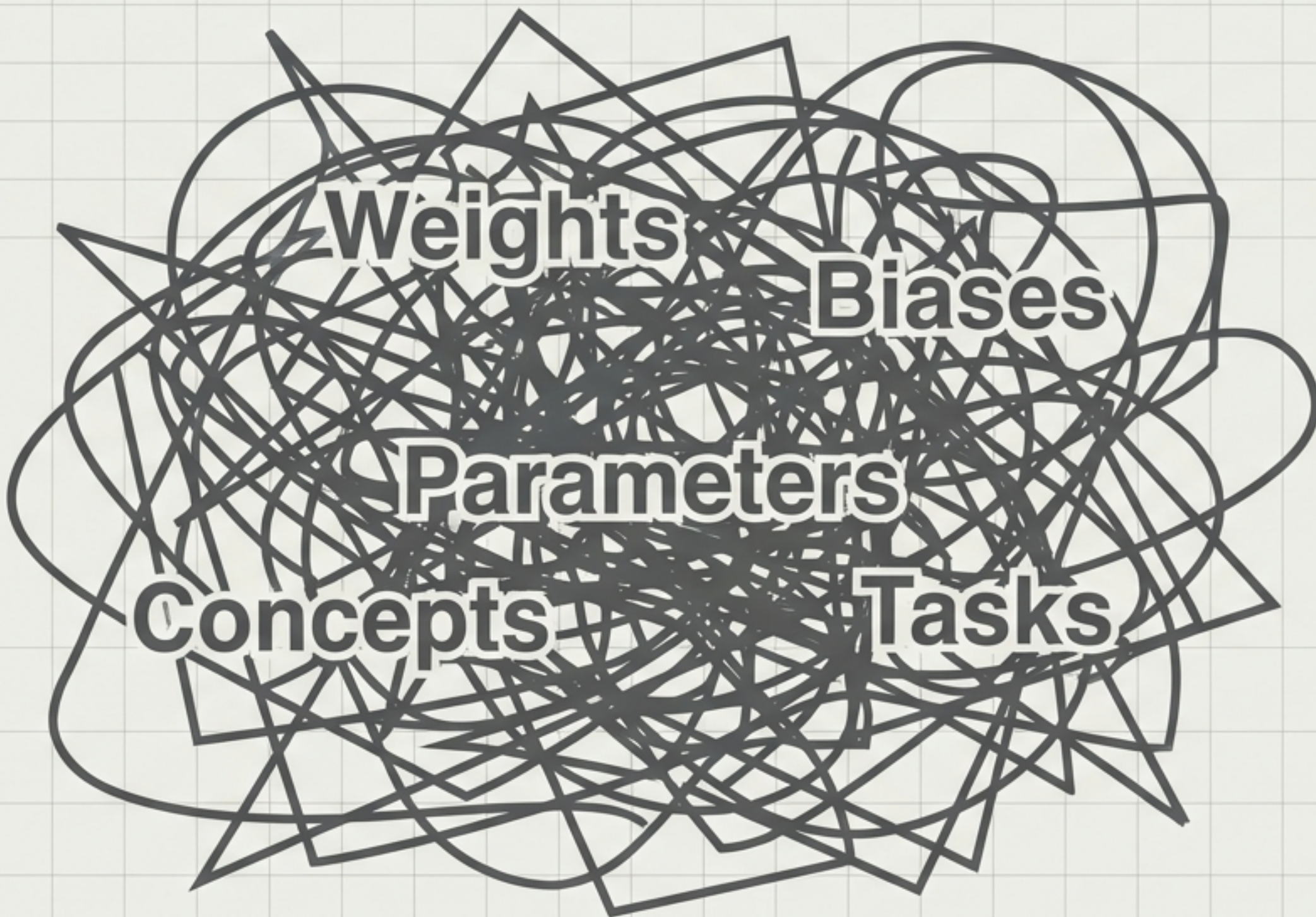
- **KEY ADVANTAGE:** Memory Safety without Garbage Collection.
- **MECHANISM:** Ownership & Borrowing rules prevent data races at compile time.
- **ADOPTION STATUS:** Entering Linux Kernel (Drivers), Android, AWS.
- **REALITY CHECK:** Rust works in concert with C/Assembly, not as a total replacement yet.

"Fearless concurrency by design."

CASE STUDY: AuraOS

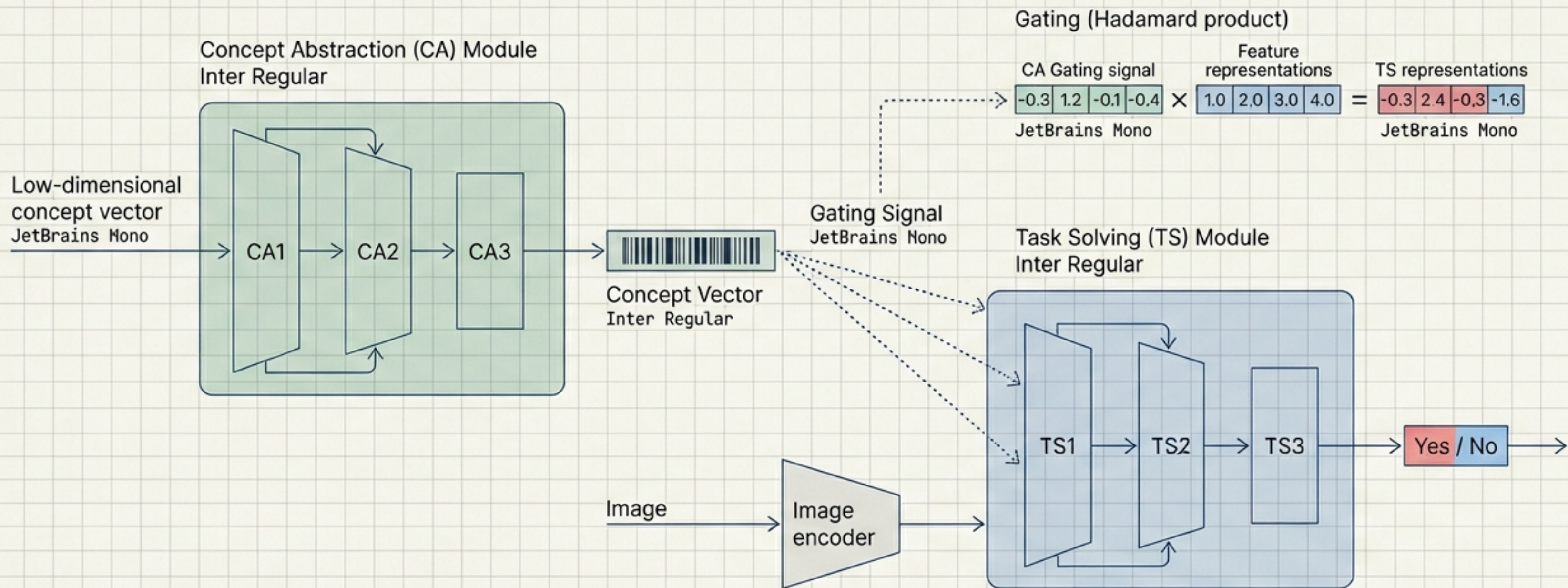


THE BLACK BOX PROBLEM

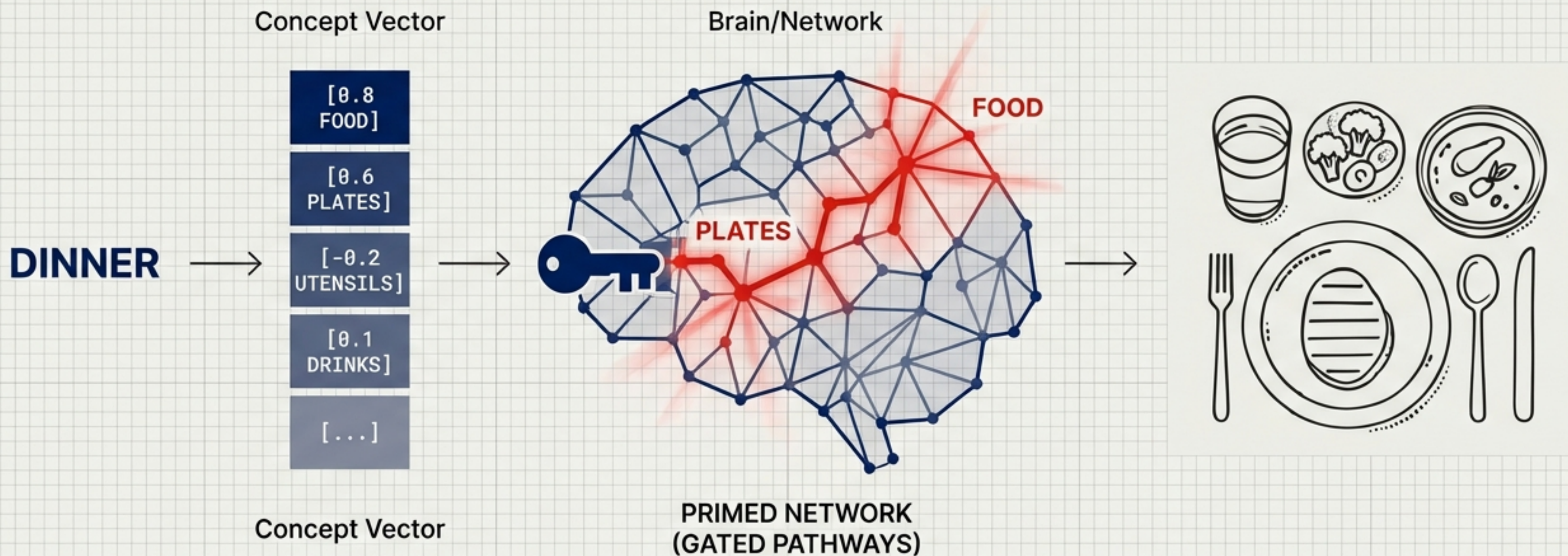


- Current Limitation: Deep Neural Networks entangle “Knowledge” with “Function”.
- The Consequence: We cannot separate the concept of an object from the task of recognizing it.
- The Goal: Decoupling Concept Abstraction (The Idea) from Task Solving (The Action).

CATS Net Architecture

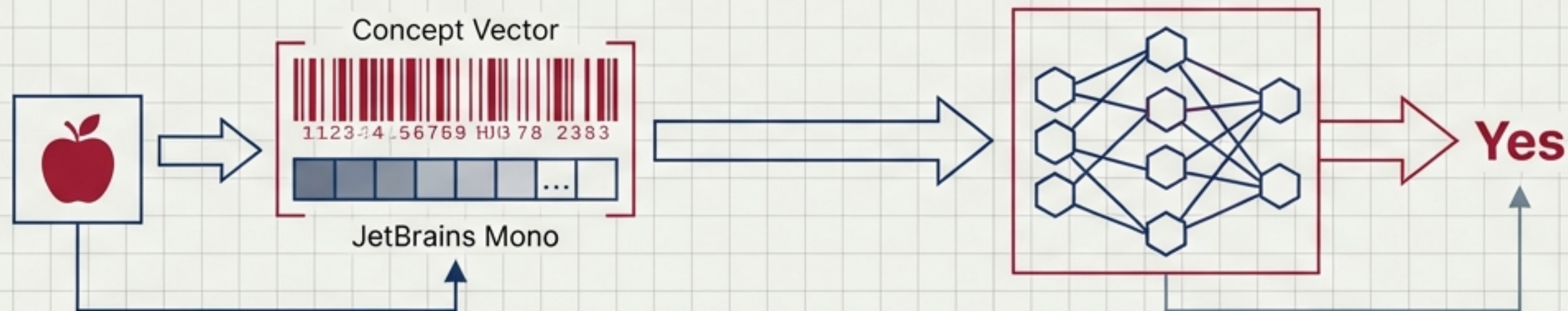


MECHANISM: THE DINNER EXAMPLE



The concept vector '**gates**' the neural pathways, reconfiguring the brain to perceive specific features before the image is even seen.

LEARNING BY COMMUNICATION



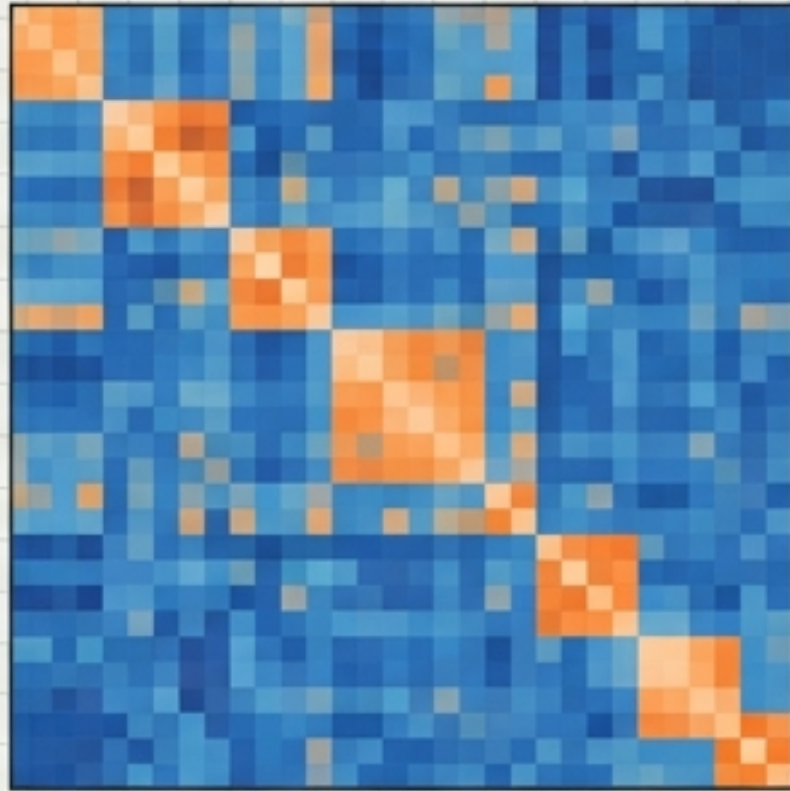
Zero-Shot Knowledge Transfer.

The student learns to recognize the object by receiving the concept vector, without weight retraining.

HUMAN ALIGNMENT

Validating the Architecture against Biological Data

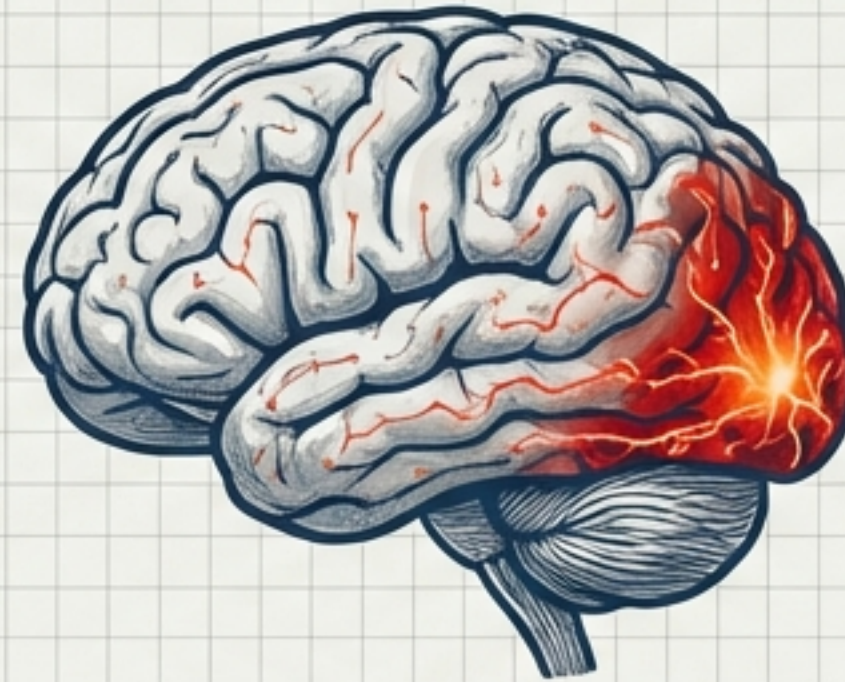
Model RDM



Similarity Matrix



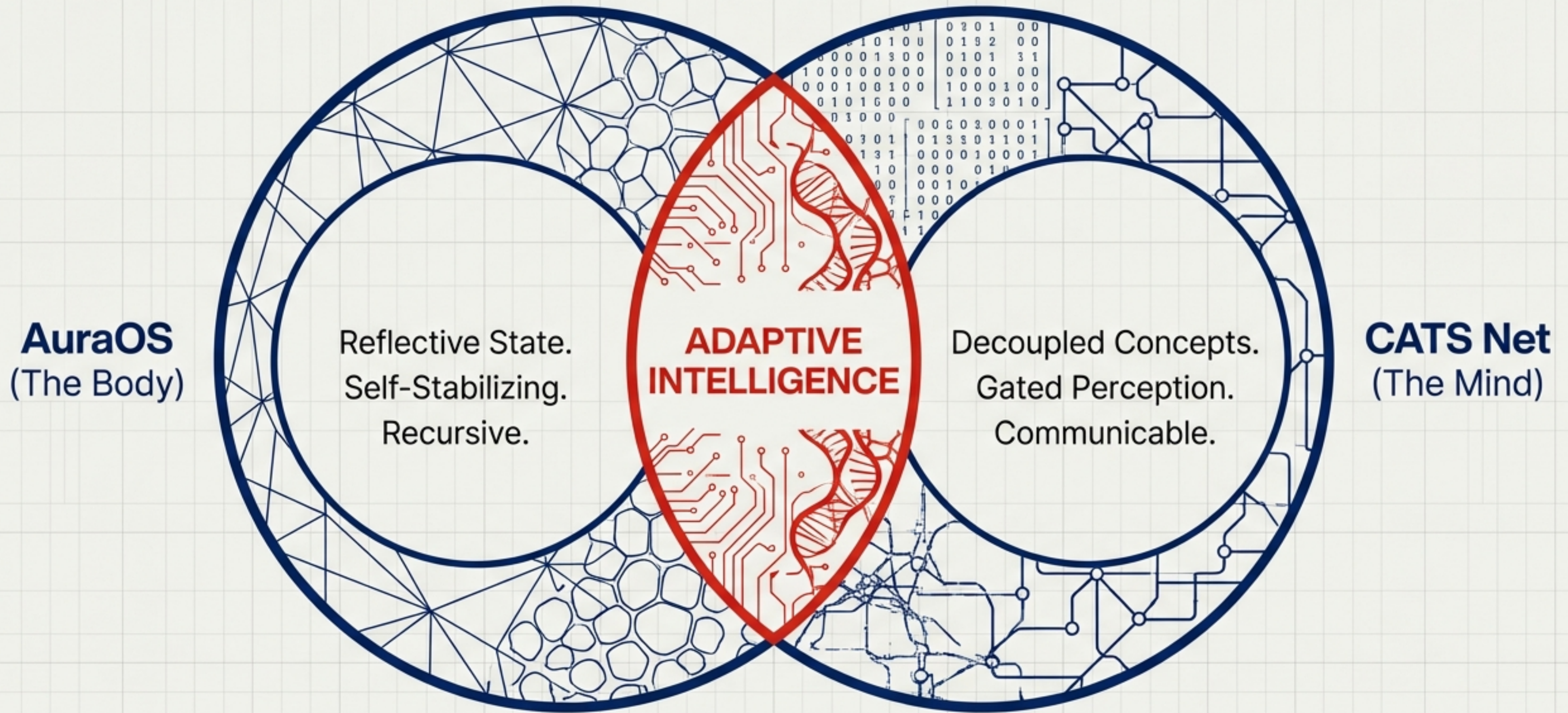
Human VOTC



Brain Scan

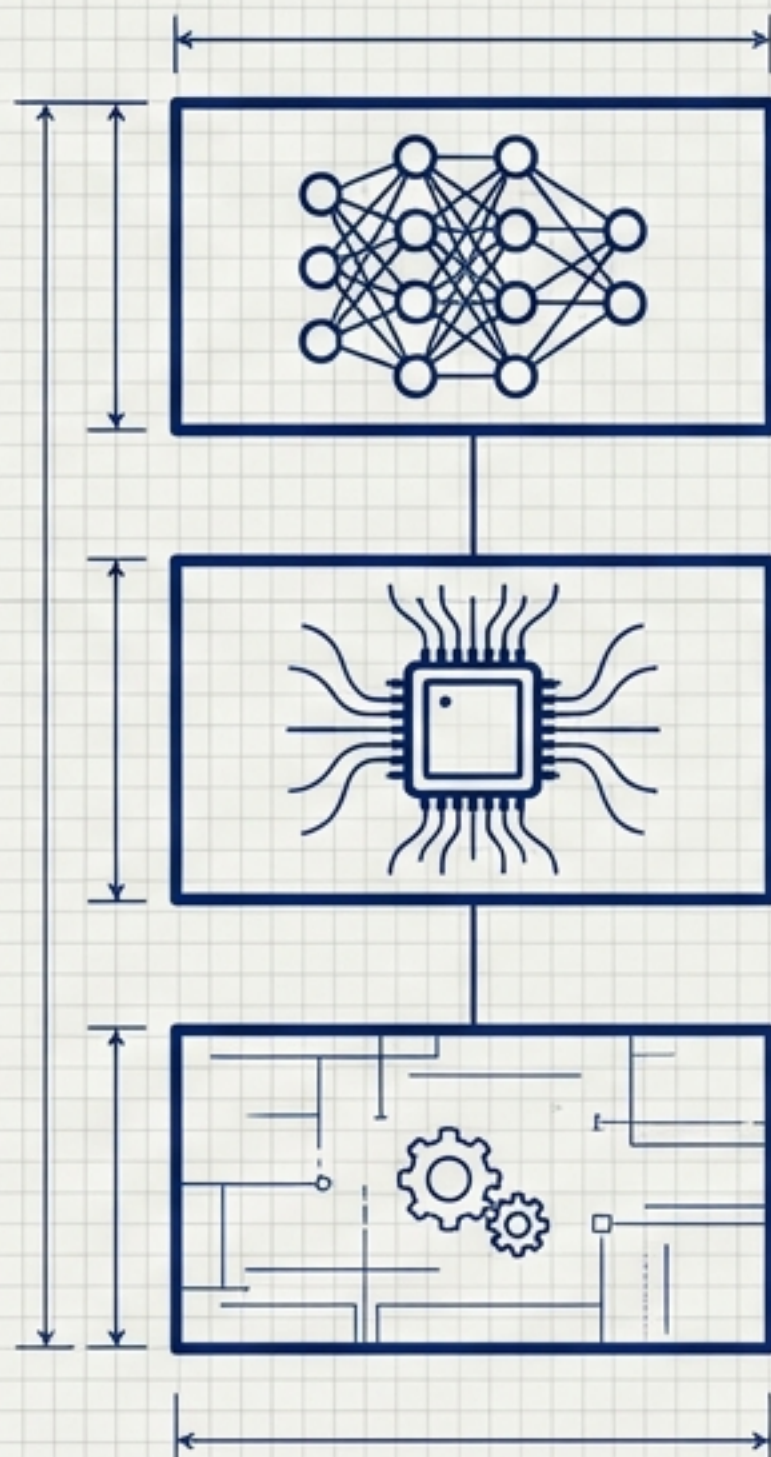
- CATS Net Concept Layer correlates with Human Visual Cortex (VOTC).
- Gating Mechanism mirrors Human Semantic Control Network.

THE COMMON THREAD



Both architectures reject rigid, linear hierarchies in favor of systems that can reflect on their parameters and adapt to context.

TOWARD RELIABLE INTELLIGENCE



CATS NET

Interpretable Cognition
Inter Regular

AuraOS

Adaptive System Logic
Inter Regular

RUST / MICROKERNEL

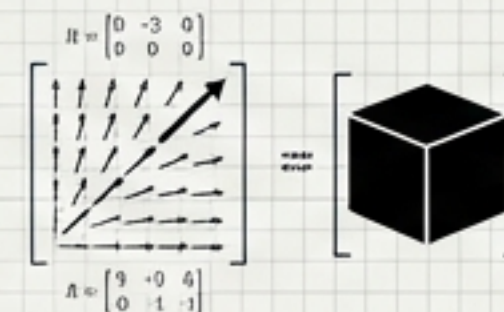
Memory Safe Foundation



SAFETY: Crash-resistant infrastructure.



INTERPRETABILITY:
Concepts are distinct
vectors, not black boxes.



COMPATIBILITY: Systems that think
and organize like humans.

Both architectures reject rigid, linear hierarchies in favor of systems that can reflect on their parameters and adapt to context.

SOURCES

1. Systems Programming & Architecture: Comparative Analysis of Monolithic, Microkernel, Exokernel, and AuraOS.
2. Guo et al. (2026). "A neural network for modeling human concept formation, understanding, and communication." Nature Computational Science.
3. Image Credits: Public Domain / Nature Computational Science.